

Tianxiang Wu

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

Bachelor of Science in Mathematics & Computer Science | GPA: 4.00/4.00

Aug 2022 – May 2026

Honors and Awards: Graduated with Highest Honors; Dean's List

Relevant Courses: Discrete Mathematics, Differential Equations, Abstract Linear Algebra, Numerical Analysis, Database Systems, Graph Theory, Machine Learning, Algorithms

Harvard University

Cambridge, MA

Master of Science in Computational Science and Engineering | Admitted

Sep 2026 – May 2028

Research Interest: Generative Modeling, LLM Reasoning & Agent Systems, with applications to real-world decision-making and embodied AI

RESEARCH EXPERIENCE

Schrödinger Bridge Methods for Diffusion Models

Champaign, IL

University of Illinois Urbana-Champaign | Advisor: Ge Liu

Feb 2025 – Apr 2026

Undergraduate Research Assistant

- Gain understanding of the Schrödinger Bridge theory and its connection to diffusion models by systematically parsing the mathematical derivations from first principles, including entropy-regularized optimal transport, SDE-based path-space formulations, and the variational lower bound that links static couplings with dynamic score-matching objectives.
- Translate the theoretical framework into a working PyTorch codebase for training generative models on standard benchmarks, implementing core components such as SDE solvers, score-network architectures, and sampling procedures based on the reproduced formulations.
- Conduct comparative experiments across different prior coupling strategies and model configurations on CIFAR-10 and MNIST, evaluating their impact on sample quality and mode coverage through quantitative metrics (e.g., FID) and qualitative inspection of generated output.

C to Rust Code Translation

Champaign, IL

University of Illinois Urbana-Champaign | Advisor: Reyhaneh Jabbarvand

Aug 2024 – Nov 2025

Undergraduate Research Assistant

- Construct a multi-stage AST-level preprocessing pipeline that extracts CFGs and call dependencies from legacy C repositories, normalizing architecture-specific macros and GNU inline assembly before feeding to a fine-tuned LLM translator.
- Extend pycparser with custom handlers for non-standard C dialects and implement a tree-sitter rewriting engine that converts pointer arithmetic into Rust safe iterator patterns, using region-based alias analysis to preserve original memory semantics.
- Develop a differential fuzzing harness with cargo-fuzz and property-based oracles, validating translated Rust code against original C binaries across thousands of randomly generated inputs to enforce behavioral equivalence and memory safety guarantees.

Searching for Optimal Symplectic Maps

Champaign, IL

University of Illinois Urbana-Champaign | Advisor: Ely Kerman

Aug 2024 – Dec 2024

Researcher

- Lead the search for optimal symplectic transformations as a constrained optimization problem over low-dimensional manifolds, embedding a symplectic regularization term into a TensorFlow loss function to enforce area-preservation invariants.
- Deploy Apache Spark with custom partitioners to distribute Hamiltonian Monte Carlo trajectory computations and generating-function evaluations, reducing runtime for high-dimensional parameter sweeps through adaptive data sharding strategies.
- Build an interactive visualization suite that tracks phase-space orbits and Jacobian determinants across iterated maps, enabling side-by-side comparisons between learned neural approximations and analytically derived benchmarks to pinpoint structural degeneracies.

PUBLICATION

- Tianran Sun, Saman Dehghan, **Tianxiang Wu**, Reyhaneh Jabbarvand. “*Translating Large-Scale C Repositories to Idiomatic Safe Rust*”. Submitted to ICSE ’26, May 2026.

INTERNSHIP

Envision Digital International Pte. Ltd.

Shanghai, China

AI Algorithm Engineer Intern

Jun 2026 – Present

- Develop a ReAct-style LLM orchestration agent that dynamically constructs adaptive DAG pipelines from natural-language alerts, with each node encapsulating composable time-series primitives such as STL decomposition, seasonal differencing, and lag-cross-correlation.
- Implement a hybrid operator retriever combining BM25 lexical matching with dense vector embeddings of operator docstrings, coupled with a critical-path scheduler that adaptively parallelizes independent branches based on estimated shard cardinalities and data locality, minimizing latency across diverse workload shapes.
- Integrate a confidence-aware fuzzy resolver that triggers interactive clarification dialogues when intent confidence is low, and containerize the FastAPI service with Kubernetes HPA for horizontal scaling, ensuring robust handling of ambiguous queries and varying concurrent tenant loads.

Technical Consulting & Research, Inc.

Champaign, IL

Software Engineer Intern

Nov 2025 – Feb 2026

- Build a Flask-based compliance assessment platform for SSP and POA&M workflows, implementing backend services for evidence capture, remediation tracking, parameter validation, and role-based access control across team users.
- Design a provider-agnostic LLM integration layer to generate POA&M evidence artifacts, exposing a reusable adapter interface that simplified switching model providers while keeping prompt and output handling consistent.
- Model control and remediation data in MySQL, containerized services with Docker, and added CI checks for safer releases, improving deployment repeatability and backend reliability for a production-style internal tool.

Jiangsu Hoperun Software Co., Ltd.

Nanjing, China

Software Engineer Intern

Jun 2023 – Aug 2023

- Build a data-driven automation framework using parametrized PyTest fixtures covering OTA firmware hardware configurations, standardizing test environment replication through Dockerized nodes to cut manual regression cycles.
- Integrate Jenkins multi-branch pipelines with Git pre-commit hooks for incremental test triggering, deploying Redis-backed caching layers and MongoDB versioned collections to reduce average backend retrieval times from 600ms to under 200ms.
- Construct a containerized RESTful API regression suite that spins up ephemeral databases per test run, ensuring complete isolation between concurrent build pipelines while enriching failure diagnostics with structured JSON logging.

TEACHING EXPERIENCE

CS 225: Data Structures

Champaign, IL

Course Assistant

Feb 2024 – May 2024

- Create detailed review notes and problem summaries to help students consolidate key C++ and data structure concepts.
- Lead comprehensive exam review sessions and addressed common questions on algorithm design and code implementation.
- Assist peers in reinforcing conceptual understanding through interactive review discussions and problem-solving practice.

SKILLS

Mathematical & Theoretical Foundations: Optimal Transport, Schrödinger Bridge, SDE/ODE, Stochastic Processes, Probability Theory, Functional Analysis, Linear Algebra

Generative Modeling & Deep Learning: Diffusion Models, Flow Matching, Score-based Models, Energy-Based Models (EBM), Representation Learning, PyTorch, JAX, TensorFlow, Distributed Training

Large Language Models & Agent Systems: LLM Alignment (RLHF/DPO), Instruction Tuning, Reasoning (CoT/ToT/Test-Time Search), RAG, Tool Use/Function Calling, Multi-Agent Collaboration, LangChain, Hugging Face Transformers

Systems & Experimentation: Python, C++, Rust, Docker, Kubernetes, Distributed Computing (Spark), Experiment Tracking, pytest, Git

Research & Analysis: Benchmarking & Comparative Analysis, Scientific Visualization (Matplotlib/NumPy), Automated Testing for Research Validation